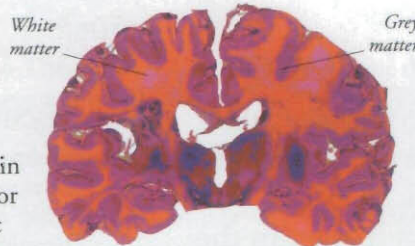
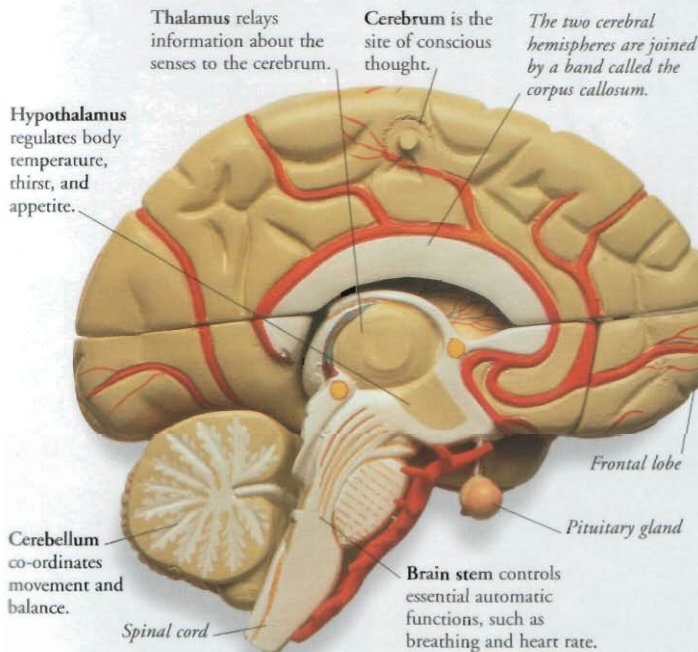


Brain

The brain is the body's control centre. Your brain enables you to think and to have a personality, and also regulates all your body processes. It has three main regions: the forebrain, the cerebellum, and the brain stem. The forebrain consists of the cerebrum (which is made up of two halves or hemispheres), the thalamus, hypothalamus, and the limbic system, which controls emotions and instinctive behaviour.



Section through brain tissue

Left and right brains

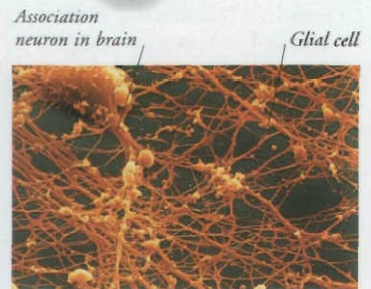
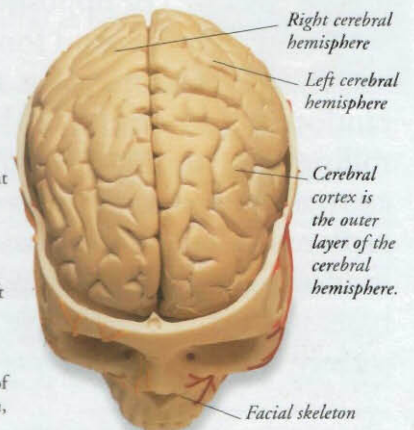
The left cerebral hemisphere controls the right side of the body, and the right cerebral hemisphere controls the left side of the body. Although both hemispheres are used for almost every activity, each hemisphere has its own specialist skills. In most people, the left hemisphere is involved in spoken and written language, mathematical ability, and reasoning, while the right hemisphere controls the appreciation of art and music, insight and imagination, and shape recognition.

Brain cells

The brain consists of hundreds of billions of nerve cells. Many of these are association neurons that are constantly receiving and transmitting nerve impulses. Any one of these neurons can have links to over 1,000 other neurons, producing a complex network. The brain also contains other nerve cells, called glial cells, which hold the neurons in place.

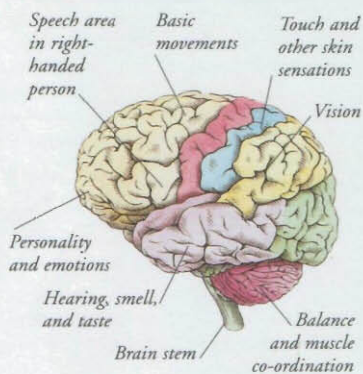
Grey and white matter

Each cerebral hemisphere has two layers. The outer layer, the cerebral cortex, consists of grey matter containing cell bodies of neurons that form a communication network. The inner layer, or white matter, consists of nerve fibres that link the cerebral cortex to the other parts of the brain.



Brain areas

Certain areas of the cerebrum are involved with particular body functions. These areas can be highlighted on a brain map. Motor areas of the brain, such as the speech and basic movement areas, send out instructions to control voluntary movement. Sensory areas, such as the hearing, taste, smell, touch, and vision areas, receive information from sensory receptors around the body. Association areas, such as the frontal lobe, deal with thoughts, personality, and emotions, analyse experiences, and give you consciousness and awareness.



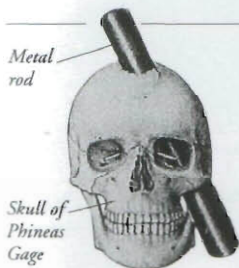
Brain waves

The brain's neurons are constantly sending out and receiving nerve impulses. This process produces electrical signals that can be detected using a machine called an electroencephalograph (EEG). Electrodes linked to the EEG can be attached to a person's scalp in order to record the brain's electrical activities as a series of patterns called brain waves.



Sleep and dreams

As you sleep, you move repeatedly between phases of light REM (rapid eye movement) sleep and phases of deeper NREM sleep. These shifts can be detected using an EEG.



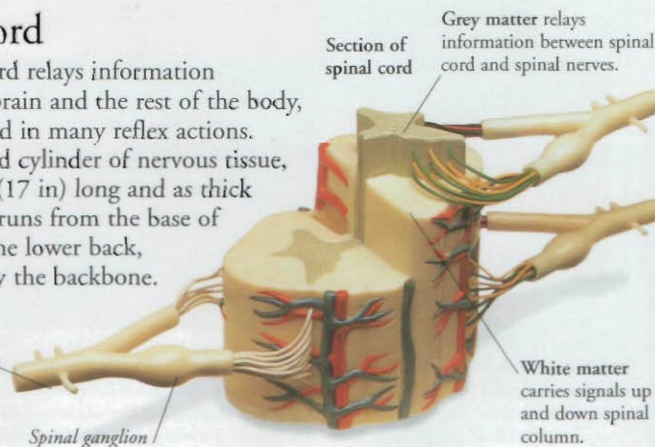
Personality

The frontal lobe of the brain plays a major role in deciding personality. This was shown by the case of an American worker called Phineas Gage. In 1848, an accident sent a metal rod through Gage's cheek and frontal lobe. He survived but his personality changed from being friendly to being aggressive.

Spinal cord

The spinal cord relays information between the brain and the rest of the body, and is involved in many reflex actions. It is a flattened cylinder of nervous tissue, about 43 cm (17 in) long and as thick as a finger. It runs from the base of the brain to the lower back, surrounded by the backbone.

Spinal nerve relays nerve impulses to and from all parts of body.



Pierre Paul Broca

French anatomist and surgeon Pierre Paul Broca (1824–80) demonstrated that a specific region of the brain controlled a particular body function. Broca found that a small area co-ordinated the muscles in the mouth and throat that produce speech. This area is now called Broca's area, or the speech area. Broca made his discovery when treating a patient who could not talk after damage to part of his brain.



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